

25: PostgreSQL Mastery Capstone Program

12-Week Intensive — From SQL Basics to Production Expert

What Is This Program?

The Capstone Program is a structured, 12-week daily study curriculum that takes you from SQL fundamentals to production-grade PostgreSQL expertise. It is the glue that holds together all 24 other chapters in this repository, providing a learning path, daily exercises, mock interviews, and a final project.

Commitment: 1-2 hours/day, 5 days/week

Outcome: Interview-ready PostgreSQL mastery + 10-project portfolio

Program Files

File	Week	Phase	Topic
00_capstone_overview.md	—	Overview	Prerequisites, outcomes, evaluation criteria
01_week1_foundations.md	1	Phase 1	Setup, fundamentals, basic SQL
02_week2_sql_mastery.md	2	Phase 1	JOINS, aggregations, subqueries
03_week3_advanced_sql.md	3	Phase 2	Window functions, CTEs, advanced patterns
04_week4_postgresql_core.md	4	Phase 2	JSONB, PL/pgSQL, custom types, domains
05_week5_database_design.md	5	Phase 2	Normalization, schema design, ER modeling
06_week6_indexes_optimization.md	6	Phase 2	All index types, EXPLAIN ANALYZE, anti-patterns
07_week7_internals.md	7	Phase 3	Transactions, MVCC, locking, UPSERT
08_week8_advanced_features.md	8	Phase 3	Partitioning, FTS, pg_trgm, extensions
09_week9_production.md	9	Phase 3	Monitoring, security, backup/recovery
10_week10_replication_ha.md	10	Phase 4	Replication, Patroni, PgBouncer, HA design
11_week11_interview_prep.md	11	Phase 4	Mock interviews, coding under pressure
12_week12_capstone_project.md	12	Phase 4	Build complete project, peer review, assessment
assessment_rubric.md	—	Evaluation	Detailed grading criteria across 5 dimensions

Learning Phases

PHASE 1: FOUNDATIONS (Weeks 1-2)	_____
Week 1:	Environment setup, data types, basic SELECT
Week 2:	JOINS, aggregations, subqueries, set operations
PHASE 2: POSTGRESQL MASTERY (Weeks 3-6)	_____
Week 3:	Window functions, CTEs, advanced patterns
Week 4:	JSONB, PL/pgSQL, domains, generated columns
Week 5:	Normalization, design patterns, ER modeling
Week 6:	B-Tree/GIN/BRIN indexes, EXPLAIN ANALYZE
PHASE 3: INTERNALS & OPERATIONS (Weeks 7-9)	_____
Week 7:	MVCC, isolation levels, locking, UPSERT
Week 8:	Partitioning, full-text search, extensions
Week 9:	Security, monitoring, backup, configuration
PHASE 4: ARCHITECTURE & CAREER (Weeks 10-12)	_____
Week 10:	Streaming replication, Patroni, PgBouncer
Week 11:	Interview preparation, mock sessions
Week 12:	Capstone project, peer review, final assessment

Repository Chapter Map

Weeks	Chapters in This Repo
1-2	01_Fundamentals, 02_SQL_Basics, 03_Intermediate_SQL
3	04_Advanced_SQL
4	05_PostgreSQL_Core
5	06_Database_Design
6	07_Indexes, 08_Query_Optimization
7	09_Transactions_Concurrency, 10_PostgreSQL_Internals
8	11_Advanced_PostgreSQL
9	12_Production_PostgreSQL, 14_Security, 15_Backup_Recovery
10	13_Replication_HA
11	19_Interview_Questions, 20_Interview_Tasks, 21_System_Design
12	22_Projects

Skills You Will Have After Completion

SQL Skills

- Complex window functions (OVER, PARTITION BY, frame clauses)
- Recursive CTEs for hierarchies
- JSONB querying with GIN indexes
- Full-text search with tsvector and tsquery
- Advanced aggregations with FILTER and GROUPING SETS

PostgreSQL Skills

- PL/pgSQL functions with full error handling
- Trigger design for data integrity
- Table partitioning (range, list, hash)
- Row-Level Security policies
- MVCC and isolation levels

Operations Skills

- EXPLAIN ANALYZE for query optimization
- Index strategy (all 6 types)
- Streaming and logical replication
- Backup and point-in-time recovery
- Connection pooling with PgBouncer
- High-availability with Patroni

Career Skills

- Answering any interview question in 2 minutes
- Live schema design under pressure
- Portfolio of 10 complete projects
- Production incident response framework

Evaluation

Your work is evaluated across 5 dimensions (see [assessment rubric.md](#)):

Dimension	Weight
Schema Design	25%
SQL Query Quality	25%
PL/pgSQL Functions/Triggers	20%
Performance Optimization	20%
Documentation/Communication	10%

Score bands: Expert (90+), Proficient (75-89), Competent (60-74), Developing (<60)

Getting Started

1. Read [00_capstone_overview.md](#) completely.
2. Ensure PostgreSQL 16 is installed and you can connect via `psql`.
3. Create a dedicated database: `CREATE DATABASE capstone;`
4. Open [01_week1_foundations.md](#) and start Day 1.

5. Track your progress using the weekly self-assessment template.
 6. Post your completed capstone project to GitHub.
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Quick Links by Goal

I want to pass a technical SQL interview this month: → Weeks 3, 6, 7, 11 are highest priority

I want to learn PostgreSQL-specific features: → Weeks 4, 7, 8, 10

I want to understand production database operations: → Weeks 6, 7, 9, 10

I want to build projects for my portfolio: → Complete all weeks, then 22_Projects/